

Storage management: Logical Volume Management (LVM) in Linux. Common hardware issues and troubleshooting techniques. Practical (2 hours/day): Configure LVM and manage partitions in Linux. Simulate hardware failures and practice troubleshooting.

Tasks

1. Use the Linux terminal to create a new partition on a virtual disk using the fdisk tool, then format it as ext4 and mount it to /mnt/music.

Answer :-

```
diksha@diksha-VMware-Virtual-Platform: ~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0        7:0    0   74M  1 loop /snap/core22/2292
loop1        7:1    0    4K  1 loop /snap/bare/5
loop2        7:2    0  66.8M  1 loop /snap/core24/1643
loop3        7:3    0  66.8M  1 loop /snap/core24/2124
loop4        7:4    0 251.7M  1 loop /snap/firefox/7766
loop5        7:5    0 531.4M  1 loop /snap/gnome-42-2204/247
loop6        7:6    0  16.5M  1 loop /snap/firmware-updater/226
loop7        7:7    0  18.5M  1 loop /snap/firmware-updater/210
loop8        7:8    0 614.5M  1 loop /snap/gnome-46-2404/164
loop9        7:9    0  402M  1 loop /snap/alsa-2404/1839
loop10       7:10   0  91.7M  1 loop /snap/gtk-common-themes/1535
loop11       7:11   0 531.5M  1 loop /snap/gnome-42-2204/263
loop12       7:12   0  11.8M  1 loop /snap/snap-store/1390
loop13       7:13   0  10.8M  1 loop /snap/snap-store/1270
loop14       7:14   0  50.1M  1 loop /snap/snapd/27591
loop15       7:15   0  50.3M  1 loop /snap/snapd/27738
loop16       7:16   0   576K  1 loop /snap/snapd-desktop-integration/343
loop17       7:17   0   828K  1 loop /snap/snapd-desktop-integration/391
sda          8:0    0   20G  0 disk 
├─sda1       8:1    0    1M  0 part 
├─sda2       8:2    0   20G  0 part /
└─sdb        8:16   0    5G  0 disk 
   └─sdb1     8:17   0    5G  0 part 
      └─media_vg-photos_lv 252:0   0   1.5G  0 lvm 
sdc          8:32   0    5G  0 disk 
sdd          8:48   0   20G  0 disk 
sr0         11:0    1 1024M  0 rom  
diksha@diksha-VMware-Virtual-Platform:~$ sudo fdisk /dev/sdd
[sudo] password for diksha:

Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xf72d1f15.

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p):
```

```
Oct 1 08:13
diksha@diksha-VMware-Virtual-Platform: ~
└─media_vg-photos_lv 252:0    0   1.5G  0 lvm
sdc      8:32    0    5G  0 disk
sdd      8:48    0   20G  0 disk
sr0      11:0    1  1024M  0 rom
diksha@diksha-VMware-Virtual-Platform:~$ sudo fdisk /dev/sdd
[sudo] password for diksha:
Welcome to fdisk (util-linux 2.39.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xf72d1f15.

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-41943039, default 2048):
Last sector, +/-sectors or +/-size[K,M,G,T,P] (2048-41943039, default 41943039):

Created a new partition 1 of type 'Linux' and of size 20 GiB.

Command (m for help): p
Disk /dev/sdd: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xf72d1f15

   Device   Boot  Start      End  Sectors  Size Id Type
/dev/sdd1               2048 41943039 41940992 20G  83 Linux

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
diksha@diksha-VMware-Virtual-Platform:~$ lsblk
```

```
Oct 1 08:13
diksha@diksha-VMware-Virtual-Platform: ~
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                               7:0      0   74M  1 loop /snap/core22/2292
loop1                               7:1      0    4K  1 loop /snap/bare/5
loop2                               7:2      0   66.8M 1 loop /snap/core24/1643
loop3                               7:3      0   66.8M 1 loop /snap/core24/2124
loop4                               7:4      0  251.7M 1 loop /snap/firefox/7766
loop5                               7:5      0  531.4M 1 loop /snap/gnome-42-2204/247
loop6                               7:6      0   16.5M 1 loop /snap/firmware-updater/226
loop7                               7:7      0   18.5M 1 loop /snap/firmware-updater/210
loop8                               7:8      0  614.5M 1 loop /snap/gnome-46-2404/164
loop9                               7:9      0   402M  1 loop /snap/mesa-2404/1839
loop10                              7:10     0   91.7M 1 loop /snap/gtk-common-themes/1535
loop11                              7:11     0  531.5M 1 loop /snap/gnome-42-2204/263
loop12                              7:12     0   11.8M 1 loop /snap/snap-store/1390
loop13                              7:13     0   10.8M 1 loop /snap/snap-store/1270
loop14                              7:14     0   50.1M 1 loop /snap/snapd/27591
loop15                              7:15     0   50.3M 1 loop /snap/snapd/27738
loop16                              7:16     0   576K  1 loop /snap/snapd-desktop-integration/343
loop17                              7:17     0   828K  1 loop /snap/snapd-desktop-integration/391
sda                                  8:0      0   20G  0 disk
├─sda1                               8:1      0    1M  0 part
├─sda2                               8:2      0   20G  0 part /
├─sdb                                8:16     0    5G  0 disk
├─sdb1                              8:17     0    5G  0 part
│   └─media_vg-photos_lv 252:0    0   1.5G  0 lvm
└─sdc                                8:32     0    5G  0 disk
    ├─sdd                           8:48     0   20G  0 disk
    └─sdd1                           8:49     0   20G  0 part
sr0                                  11:0     1  1024M  0 rom
diksha@diksha-VMware-Virtual-Platform:~$ sudo partprobe /dev/sdd
diksha@diksha-VMware-Virtual-Platform:~$ sudo mkfs.ext4 /dev/sdd1
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 5242624 4k blocks and 1310720 inodes
Filesystem UUID: 30ff6108-96d5-497e-83b0-39f409a57048
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done
diksha@diksha-VMware-Virtual-Platform:~$ sudo mkdir -p /mnt/music
diksha@diksha-VMware-Virtual-Platform:~$ sudo mount /dev/sdd1 /mnt/music
```

```
Oct 1 08:13
diksha@diksha-VMware-Virtual-Platform: ~
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done

diksha@diksha-VMware-Virtual-Platform:~$ sudo mkdir -p /mnt/music
diksha@diksha-VMware-Virtual-Platform:~$ sudo mount /dev/sdd1 /mnt/music
diksha@diksha-VMware-Virtual-Platform:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                               7:0      0   74M  1 loop /snap/core22/2292
loop1                               7:1      0    4K  1 loop /snap/bare/5
loop2                               7:2      0  66.8M  1 loop /snap/core24/1643
loop3                               7:3      0  66.8M  1 loop /snap/core24/2124
loop4                               7:4      0 251.7M  1 loop /snap/firefox/7766
loop5                               7:5      0 531.4M  1 loop /snap/gnome-42-2204/247
loop6                               7:6      0  16.5M  1 loop /snap/firmware-updater/226
loop7                               7:7      0  18.5M  1 loop /snap/firmware-updater/210
loop8                               7:8      0 614.5M  1 loop /snap/gnome-46-2404/164
loop9                               7:9      0  402M  1 loop /snap/alsa-2404/1839
loop10                              7:10     0   91.7M  1 loop /snap/gtk-common-themes/1535
loop11                              7:11     0 531.5M  1 loop /snap/gnome-42-2204/263
loop12                              7:12     0  11.8M  1 loop /snap/snap-store/1390
loop13                              7:13     0  10.8M  1 loop /snap/snap-store/1270
loop14                              7:14     0  50.1M  1 loop /snap/snapd/27591
loop15                              7:15     0  50.3M  1 loop /snap/snapd/27738
loop16                              7:16     0  576K  1 loop /snap/snapd-desktop-integration/343
loop17                              7:17     0  828K  1 loop /snap/snapd-desktop-integration/391
sda                                  8:0      0    20G  0 disk
├─sda1                              8:1      0    1M  0 part
├─sda2                              8:2      0    20G  0 part /
├─sdb                               8:16     0    5G  0 disk
├─sdb1                              8:17     0    5G  0 part
│   └─media_vg-photos_lv 252:0     0   1.5G  0 lvm
sdc                                  8:32     0    5G  0 disk
sdd                                  8:48     0    20G  0 disk
└─sdd1                              8:49     0    20G  0 part /mnt/music
sr0                                  11:0     1 1024M  0 rom

diksha@diksha-VMware-Virtual-Platform:~$ df -h /mnt/music
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdd1        20G  24K   19G   1% /mnt/music

diksha@diksha-VMware-Virtual-Platform:~$ sudo touch /mnt/music/test.txt
diksha@diksha-VMware-Virtual-Platform:~$ ls -l /mnt/music
total 16
drwx----- 2 root root 16384 Oct 1 08:11 lost+found
-rw-r--r-- 1 root root    0 Oct 1 08:12 test.txt

diksha@diksha-VMware-Virtual-Platform:~$
```

2. Set up a Logical Volume Management (LVM) group named 'media_vg' with two physical volumes, then create a logical volume 'photos_lv' of 1GB and mount it to /mnt/photos.

Hint: Use pvcreate, vgcreate, lvcreate, mkfs, and mount commands in sequence.

Answer :-

```
diksha@diksha-VMware-Virtual-Platform: ~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0   66.8M 1 loop /snap/core24/1643
loop1       7:1      0    4K   1 loop /snap/bare/5
loop2       7:2      0   66.8M 1 loop /snap/core24/2124
loop3       7:3      0    74M   1 loop /snap/core22/2292
loop4       7:4      0  251.7M 1 loop /snap/firefox/7766
loop5       7:5      0   18.5M 1 loop /snap/firmware-updater/210
loop6       7:6      0   16.5M 1 loop /snap/firmware-updater/226
loop7       7:7      0  531.4M 1 loop /snap/gnome-42-2204/247
loop8       7:8      0  531.5M 1 loop /snap/gnome-42-2204/263
loop9       7:9      0  614.5M 1 loop /snap/gnome-46-2404/164
loop10      7:10     0   91.7M 1 loop /snap/gtk-common-themes/1535
loop11      7:11     0   402M   1 loop /snap/mesa-2404/1839
loop12      7:12     0   11.8M   1 loop /snap/snap-store/1390
loop13      7:13     0   10.8M   1 loop /snap/snap-store/1270
loop14      7:14     0   50.1M   1 loop /snap/snapd/27591
loop15      7:15     0   828K   1 loop /snap/snapd-desktop-integration/391
loop16      7:16     0   50.3M   1 loop /snap/snapd/27738
loop17      7:17     0   576K   1 loop /snap/snapd-desktop-integration/343
sda         8:0      0    20G   0 disk
├─sda1      8:1      0     1M   0 part
└─sda2      8:2      0    20G   0 part /
sdb         8:16     0     5G   0 disk
sdc         8:32     0     5G   0 disk
sr0        11:0     1  1024M   0 rom

diksha@diksha-VMware-Virtual-Platform: ~$ sudo pvcreate /dev/sdb /dev/sdc
Physical volume "/dev/sdb" successfully created.
Physical volume "/dev/sdc" successfully created.
diksha@diksha-VMware-Virtual-Platform: ~$ sudo pvs
PV          VG Fmt Attr PSize PFree
/dev/sdb    lvm2 --- 5.00g 5.00g
/dev/sdc    lvm2 --- 5.00g 5.00g
diksha@diksha-VMware-Virtual-Platform: ~$ sudo vgcreate media_vg /dev/sdb /dev/sdc
Volume group "media_vg" successfully created
diksha@diksha-VMware-Virtual-Platform: ~$ sudo vgs
VG          #PV #LV #SN Attr   VSize VFree
media_vg    2   0   0 wz--n- 9.99g 9.99g
diksha@diksha-VMware-Virtual-Platform: ~$ sudo lvcreate -L 1G -n photos_lv media_vg
Logical volume "photos_lv" created.
diksha@diksha-VMware-Virtual-Platform: ~$ sudo lvs
LV          VG Attr      LSize Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert
photos_lv   media_vg -wi-a----- 1.00g
diksha@diksha-VMware-Virtual-Platform: ~$ sudo mkfs.ext4 /dev/media_vg/photos_lv
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem 1.47.0 (5-Feb-2023)
```

```
diksha@diksha-VMware-Virtual-Platform: ~$ sudo mkdir -p /mnt/photos
diksha@diksha-VMware-Virtual-Platform: ~$ sudo mount /dev/media_vg/photos_lv /mnt/photos
diksha@diksha-VMware-Virtual-Platform: ~$ df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                     387M    2.2M  385M   1% /run
/dev/sda2                 20G     13G   6.1G  68% /
tmpfs                     1.9G      0   1.9G   0% /dev/shm
tmpfs                     5.0M     8.0K   5.0M   1% /run/lock
tmpfs                     387M    120K  387M   1% /run/user/1000
/dev/mapper/media_vg-photos_lv 974M     24K  907M   1% /mnt/photos

diksha@diksha-VMware-Virtual-Platform: ~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0      0   66.8M 1 loop /snap/core24/1643
loop1       7:1      0    4K   1 loop /snap/bare/5
loop2       7:2      0   66.8M 1 loop /snap/core24/2124
loop3       7:3      0    74M   1 loop /snap/core22/2292
loop4       7:4      0  251.7M 1 loop /snap/firefox/7766
loop5       7:5      0   18.5M 1 loop /snap/firmware-updater/210
loop6       7:6      0   16.5M 1 loop /snap/firmware-updater/226
loop7       7:7      0  531.4M 1 loop /snap/gnome-42-2204/247
loop8       7:8      0  531.5M 1 loop /snap/gnome-42-2204/263
loop9       7:9      0  614.5M 1 loop /snap/gnome-46-2404/164
loop10      7:10     0   91.7M 1 loop /snap/gtk-common-themes/1535
loop11      7:11     0   402M   1 loop /snap/mesa-2404/1839
loop12      7:12     0   11.8M   1 loop /snap/snap-store/1390
loop13      7:13     0   10.8M   1 loop /snap/snap-store/1270
loop14      7:14     0   50.1M   1 loop /snap/snapd/27591
loop15      7:15     0   828K   1 loop /snap/snapd-desktop-integration/391
loop16      7:16     0   50.3M   1 loop /snap/snapd/27738
loop17      7:17     0   576K   1 loop /snap/snapd-desktop-integration/343
sda         8:0      0    20G   0 disk
├─sda1      8:1      0     1M   0 part
└─sda2      8:2      0    20G   0 part /
sdb         8:16     0     5G   0 disk
└─media_vg-photos_lv 252:0     0    1G   0 lvm  /mnt/photos
sdc         8:32     0     5G   0 disk
sr0        11:0     1  1024M   0 rom

diksha@diksha-VMware-Virtual-Platform: ~$ sudo touch /mnt/photos/test.txt
diksha@diksha-VMware-Virtual-Platform: ~$ ls -l /mnt/photos
total 16
drwxr-xr-x 2 root root 16384 Oct 1 07:45 test+found
-rw-r--r-- 1 root root    0 Oct 1 07:45 test.txt
diksha@diksha-VMware-Virtual-Platform: ~$
```

3. Simulate a hardware failure by unmounting a mounted partition and deleting its mount directory, then use Linux commands to diagnose and recover access to the data.

Hint: Try using lsblk, mount, and dmesg to help troubleshoot.

Answer :-

```
Oct 1 08:17
diksha@diksha-VMware-Virtual-Platform: ~
loop14 7:14 0 50.1M 1 loop /snap/snapd/27591
loop15 7:15 0 50.3M 1 loop /snap/snapd/27738
loop16 7:16 0 576K 1 loop /snap/snapd-desktop-integration/343
loop17 7:17 0 828K 1 loop /snap/snapd-desktop-integration/391
sda 8:0 0 20G 0 disk
├─sda1 8:1 0 1M 0 part
├─sda2 8:2 0 20G 0 part /
├─sdb 8:16 0 5G 0 disk
├─sdb1 8:17 0 5G 0 part
├─media_vg-photos_lv 252:0 0 1.5G 0 lvm
├─sdc 8:32 0 5G 0 disk
├─sdd 8:48 0 20G 0 disk
├─sdd1 8:49 0 20G 0 part
└─sr0 11:0 1 1024M 0 rom
diksha@diksha-VMware-Virtual-Platform: ~$ sudo dmesg | tail -20
[ 14.007313] NET: Registered PF_QIPCRTR protocol family
[ 14.527531] set_capacity_and_notify: 0 callbacks suppressed
[ 14.527585] loop18: detected capacity change from 0 to 8
[ 14.956777] kauditd_printk_skb: 147 callbacks suppressed
[ 14.956806] audit: type=1400 audit(1790822334.890:159): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="/usr/lib/sn
apd/snap-confine" pid=1364 comm="apparmor_parser"
[ 15.591048] piix4_smbus 0000:00:07.3: SMBus Host Controller not enabled!
[ 16.068005] e1000: ens33 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: None
[ 16.311755] audit: type=1400 audit(1790822336.245:160): apparmor="DENIED" operation="capable" class="cap" profile="/usr/sbin/cupsd" pid=1499 comm="cupsd" capability=12 capname="net
_admin"
[ 16.763792] RAPL PMU: API unit is 2^-32 Joules, 0 fixed counters, 10737418240 ms ovfl timer
[ 17.082155] intel_rapl_msr: PL4 support detected.
[ 22.570507] rfkill: input handler disabled
[ 30.840119] systemd-journald[379]: /var/log/journal/d332fc31760d44c7ba11c53cc308f024/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 31.536722] rfkill: input handler enabled
[ 34.953942] rfkill: input handler disabled
[ 39.424040] systemd-journald[379]: Time jumped backwards, rotating.
[ 40.814546] input: VMware DnD Uinput pointer as /devices/virtual/input/input6
[ 152.884939] sdd: sdd1
[ 186.992502] sdd: sdd1
[ 207.640057] EXT4-fs (sdd1): mounted filesystem 30ff6108-96d5-497e-83b0-39f409a57048 r/w with ordered data mode. Quota mode: none.
[ 364.734130] EXT4-fs (sdd1): unmounting filesystem 30ff6108-96d5-497e-83b0-39f409a57048.
diksha@diksha-VMware-Virtual-Platform: ~$ sudo mount /dev/sdd1 /mnt/music
mount: /mnt/music: mount point does not exist.
dmesg(1) may have more information after failed mount system call.
diksha@diksha-VMware-Virtual-Platform: ~$ sudo mkdir -p /mnt/music
diksha@diksha-VMware-Virtual-Platform: ~$ sudo mount /dev/sdd1 /mnt/music
diksha@diksha-VMware-Virtual-Platform: ~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 74M 1 loop /snap/core22/2292
loop1 7:1 0 4M 1 loop /snap/core22/2292
loop2 7:2 0 4M 1 loop /snap/core22/2292
loop3 7:3 0 4M 1 loop /snap/core22/2292
loop4 7:4 0 4M 1 loop /snap/core22/2292
loop5 7:5 0 4M 1 loop /snap/core22/2292
loop6 7:6 0 4M 1 loop /snap/core22/2292
loop7 7:7 0 4M 1 loop /snap/core22/2292
loop8 7:8 0 4M 1 loop /snap/core22/2292
loop9 7:9 0 4M 1 loop /snap/core22/2292
loop10 7:10 0 4M 1 loop /snap/core22/2292
loop11 7:11 0 4M 1 loop /snap/core22/2292
loop12 7:12 0 4M 1 loop /snap/core22/2292
loop13 7:13 0 4M 1 loop /snap/core22/2292
loop14 7:14 0 4M 1 loop /snap/core22/2292
loop15 7:15 0 4M 1 loop /snap/core22/2292
loop16 7:16 0 4M 1 loop /snap/core22/2292
loop17 7:17 0 4M 1 loop /snap/core22/2292
loop18 7:18 0 4M 1 loop /snap/core22/2292
loop19 7:19 0 4M 1 loop /snap/core22/2292
loop20 7:20 0 4M 1 loop /snap/core22/2292
loop21 7:21 0 4M 1 loop /snap/core22/2292
loop22 7:22 0 4M 1 loop /snap/core22/2292
loop23 7:23 0 4M 1 loop /snap/core22/2292
loop24 7:24 0 4M 1 loop /snap/core22/2292
loop25 7:25 0 4M 1 loop /snap/core22/2292
loop26 7:26 0 4M 1 loop /snap/core22/2292
loop27 7:27 0 4M 1 loop /snap/core22/2292
loop28 7:28 0 4M 1 loop /snap/core22/2292
loop29 7:29 0 4M 1 loop /snap/core22/2292
loop30 7:30 0 4M 1 loop /snap/core22/2292
loop31 7:31 0 4M 1 loop /snap/core22/2292
loop32 7:32 0 4M 1 loop /snap/core22/2292
loop33 7:33 0 4M 1 loop /snap/core22/2292
loop34 7:34 0 4M 1 loop /snap/core22/2292
loop35 7:35 0 4M 1 loop /snap/core22/2292
loop36 7:36 0 4M 1 loop /snap/core22/2292
loop37 7:37 0 4M 1 loop /snap/core22/2292
loop38 7:38 0 4M 1 loop /snap/core22/2292
loop39 7:39 0 4M 1 loop /snap/core22/2292
loop40 7:40 0 4M 1 loop /snap/core22/2292
loop41 7:41 0 4M 1 loop /snap/core22/2292
loop42 7:42 0 4M 1 loop /snap/core22/2292
loop43 7:43 0 4M 1 loop /snap/core22/2292
loop44 7:44 0 4M 1 loop /snap/core22/2292
loop45 7:45 0 4M 1 loop /snap/core22/2292
loop46 7:46 0 4M 1 loop /snap/core22/2292
loop47 7:47 0 4M 1 loop /snap/core22/2292
loop48 7:48 0 4M 1 loop /snap/core22/2292
loop49 7:49 0 4M 1 loop /snap/core22/2292
loop50 7:50 0 4M 1 loop /snap/core22/2292
loop51 7:51 0 4M 1 loop /snap/core22/2292
loop52 7:52 0 4M 1 loop /snap/core22/2292
loop53 7:53 0 4M 1 loop /snap/core22/2292
loop54 7:54 0 4M 1 loop /snap/core22/2292
loop55 7:55 0 4M 1 loop /snap/core22/2292
loop56 7:56 0 4M 1 loop /snap/core22/2292
loop57 7:57 0 4M 1 loop /snap/core22/2292
loop58 7:58 0 4M 1 loop /snap/core22/2292
loop59 7:59 0 4M 1 loop /snap/core22/2292
loop60 7:60 0 4M 1 loop /snap/core22/2292
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loop38
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4. Resize the logical volume 'photos_lv' you created earlier to increase its size by 500MB, then verify the new size using the appropriate LVM and filesystem commands.

Answer :-

```
diksha@diksha-VMware-Virtual-Platform:~$
diksha@diksha-VMware-Virtual-Platform:~$ sudo lvextend -L +500M /dev/media_vg/photos_lv
Size of logical volume media_vg/photos_lv changed from 1.00 GiB (256 extents) to <1.49 GiB (381 extents).
Logical volume media_vg/photos_lv successfully resized.
diksha@diksha-VMware-Virtual-Platform:~$ sudo resize2fs /dev/media_vg/photos_lv
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/media_vg/photos_lv is mounted on /mnt/photos; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 1
The filesystem on /dev/media_vg/photos_lv is now 390144 (4k) blocks long.

diksha@diksha-VMware-Virtual-Platform:~$ sudo lvs
LV          VG          Attr      LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
photos_lv   media_vg     -wi-ao---- <1.49g

diksha@diksha-VMware-Virtual-Platform:~$ df -h /mnt/photos
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/media_vg-photos_lv 1.5G      24K    1.4G   1% /mnt/photos
diksha@diksha-VMware-Virtual-Platform:~$
diksha@diksha-VMware-Virtual-Platform:~$
```

5. A user reports that their USB drive is not detected when plugged in. List 3 Linux commands you would use to troubleshoot this issue and explain what each command checks.

Answer :-

```
Oct 1 08:18
diksha@diksha-VMware-Virtual-Platform:~$ lsusb
Bus 001 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
Bus 001 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 001 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
diksha@diksha-VMware-Virtual-Platform:~$ sudo dmesg | tail -20
[ 14.527531] set_capacity_and_notify: 8 callbacks suppressed
[ 14.527585] loop18: detected capacity change from 0 to 8
[ 14.956777] kauditd_printk_skb: 147 callbacks suppressed
[ 14.956806] audit: type=1400 audit(1790822334.890:159): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="/usr/lib/sn
apd/snap-confine" pid=1364 comm="apparmor_parser"
[ 15.591048] piix4_smbus 0000:00:07:3: SMBus Host Controller not enabled!
[ 16.068005] e1000: ens33 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: None
[ 16.311755] audit: type=1400 audit(1790822336.245:160): apparmor="DENIED" operation="capable" class="cap" profile="/usr/sbin/cupsd" pid=1499 comm="cupsd" capability=12 capname="net
_admin"
[ 16.763792] RAPL PMU: API unit is 2^-32 Joules, 0 fixed counters, 10737418240 ms ovfl timer
[ 17.082155] intel_rapl_msr: PL4 support detected.
[ 22.570507] rfkil: input handler disabled
[ 30.840119] systemd-journald[379]: /var/log/journal/d332fc31760d44c7ba11c53cc308f024/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 31.536722] rfkil: input handler enabled
[ 34.953942] rfkil: input handler disabled
[ 39.424840] systemd-journald[379]: Time jumped backwards, rotating.
[ 40.814546] input: VMware DnD Uinput pointer as /devices/virtual/input/input6
[ 152.884939] sdd: sdd1
[ 186.992502] sdd: sdd1
[ 207.640057] EXT4-fs (sdd1): mounted filesystem 30ff6108-96d5-497e-83b0-39f409a57048 r/w with ordered data mode. Quota mode: none.
[ 364.734130] EXT4-fs (sdd1): unmounting filesystem 30ff6108-96d5-497e-83b0-39f409a57048.
[ 441.031721] EXT4-fs (sdd1): mounted filesystem 30ff6108-96d5-497e-83b0-39f409a57048 r/w with ordered data mode. Quota mode: none.
diksha@diksha-VMware-Virtual-Platform:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                               7:0      0   74M  1 loop /snap/core22/2292
loop1                               7:1      0    4K  1 loop /snap/bare/5
loop2                               7:2      0   66.8M 1 loop /snap/core24/1643
loop3                               7:3      0   66.8M 1 loop /snap/core24/2124
loop4                               7:4      0   251.7M 1 loop /snap/firefox/7766
loop5                               7:5      0   531.4M 1 loop /snap/gnome-42-2204/247
loop6                               7:6      0   16.5M  1 loop /snap/firmware-updater/226
loop7                               7:7      0   18.5M  1 loop /snap/firmware-updater/219
loop8                               7:8      0   614.5M 1 loop /snap/gnome-46-2404/164
loop9                               7:9      0   402M   1 loop /snap/mesa-2404/1839
loop10                              7:10     0   91.7M  1 loop /snap/gtk-common-themes/1535
loop11                              7:11     0   531.5M 1 loop /snap/gnome-42-2204/263
loop12                              7:12     0   11.0M  1 loop /snap/snap-store/1390
loop13                              7:13     0   10.0M  1 loop /snap/snap-store/1270
loop14                              7:14     0   50.1M  1 loop /snap/snapd/27591
```

```
Oct 1 08:18
diksha@diksha-VMware-Virtual-Platform: ~
admin"
[ 16.763792] RAPL PMU: API unit is 2^-32 Joules, 0 fixed counters, 10737418240 ms ovfl timer
[ 17.082155] intel_rapl_msr: PL4 support detected.
[ 22.570507] rfkill: input handler disabled
[ 30.840119] systemd-journald[379]: /var/log/journal/d332fc31760d44c7ba11c53cc308f024/user-1000.journal: Journal file uses a different sequence number ID, rotating.
[ 31.536722] rfkill: input handler enabled
[ 34.953942] rfkill: input handler disabled
[ 39.424840] systemd-journald[379]: Time jumped backwards, rotating.
[ 40.814546] input: VMware DnD Uinput pointer as /devices/virtual/input/input6
[ 152.884939] sdd: sdd1
[ 186.992502] sdd: sdd1
[ 207.640057] EXT4-fs (sdd1): mounted filesystem 30ff6108-96d5-497e-83b0-39f409a57048 r/w with ordered data mode. Quota mode: none.
[ 364.734130] EXT4-fs (sdd1): unmounting filesystem 30ff6108-96d5-497e-83b0-39f409a57048.
[ 441.031721] EXT4-fs (sdd1): mounted filesystem 30ff6108-96d5-497e-83b0-39f409a57048 r/w with ordered data mode. Quota mode: none.
diksha@diksha-VMware-Virtual-Platform: ~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                               7:0      0   74M  1 loop /snap/core22/2292
loop1                               7:1      0    4K  1 loop /snap/bare/5
loop2                               7:2      0   66.8M  1 loop /snap/core24/1643
loop3                               7:3      0   66.8M  1 loop /snap/core24/2124
loop4                               7:4      0   251.7M  1 loop /snap/firefox/7766
loop5                               7:5      0   531.4M  1 loop /snap/gnome-42-2204/247
loop6                               7:6      0    16.5M  1 loop /snap/firmware-updater/226
loop7                               7:7      0    18.5M  1 loop /snap/firmware-updater/210
loop8                               7:8      0   614.5M  1 loop /snap/gnome-46-2404/164
loop9                               7:9      0   402M  1 loop /snap/maas-2404/1839
loop10                              7:10     0   91.7M  1 loop /snap/gtk-common-themes/1535
loop11                              7:11     0   531.5M  1 loop /snap/gnome-42-2204/263
loop12                              7:12     0   11.8M  1 loop /snap/snap-store/1390
loop13                              7:13     0   10.8M  1 loop /snap/snap-store/1270
loop14                              7:14     0   50.1M  1 loop /snap/snapd/27591
loop15                              7:15     0   50.3M  1 loop /snap/snapd/27738
loop16                              7:16     0   576K  1 loop /snap/snapd-desktop-integration/343
loop17                              7:17     0   828K  1 loop /snap/snapd-desktop-integration/391
sda                                 8:0      0    20G  0 disk
├─sda1                              8:1      0    1M  0 part
├─sda2                              8:2      0    20G  0 part /
└─sdb                               8:16     0    5G  0 disk
    └─sdb1                          8:17     0    5G  0 part
        └─media_vg-photos_lv 252:0     0   1.5G  0 lvm
sdc                                 8:32     0    5G  0 disk
sdd                                 8:48     0    20G  0 disk
└─sdd1                              8:49     0    20G  0 part /mnt/music
sr0                                 11:0     1  1024M  0 rom
diksha@diksha-VMware-Virtual-Platform: ~$
```